
PILOT DEPLOYMENT GUIDE

Pilot Deployment Guide

Prerequisites, deployment modes, and success metrics

Aurelius

AI infrastructure orchestration and cost optimization

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Scope of a first pilot

A first pilot targets Tier 1 optimization — choosing the region and start window for flexible workloads. This is the validated capability and requires no privileged access to customer infrastructure.

Cluster and queue-aware placement (Tier 2) and GPU and node-level placement (Tier 3) can be layered on later if the customer exposes the necessary scheduler and telemetry data; they are not part of the initial pilot's success criteria.

Prerequisites

The pilot can begin with a single data export and two short confirmations. The workload trace describes the shape of jobs — what kind, how long, how flexible — not their contents.

Prerequisite	Form	Purpose
Workload trace	Recent jobs, 30–90 days	Establishes the real workload mix
Compute regions	List mapped to markets	Selects the wholesale price feeds
SLA / flexibility	Delay tolerance per type	Bounds the optimizer's headroom
Current monthly GPU spend	Cost figure	Sets the magnitude of the projection

Optional inputs

Optional input	Enables
Historical energy invoices	Cross-check of price-data alignment
Queue depth / wait-time export	Cluster and queue-aware placement (Tier 2)
GPU telemetry	GPU and node-level placement (Tier 3)
Marginal-emissions data license	Carbon-aware optimization beyond CAISO

Deployment modes

A pilot proceeds through three modes of increasing engagement. Most of the evidence is produced in the first two; the third is optional and scoped to flexible workloads.

Offline replay

Aurelius re-runs the historical workload trace against historical market prices and reports the cost reduction the optimizer would have achieved. This is a read-only analysis that touches no live system and produces an initial, workload-specific projection.

Shadow mode

Aurelius records the decisions it would make against live market data — for each job, the chosen region and start window, the forecast, and the predicted savings — without executing any workload. After the settlement window closes, realized prices are compared to predicted savings, per job and per workload class. This is the primary source of credible, customer-specific economic evidence.

Controlled execution

Optionally, and only after shadow-mode results support it, Aurelius acts on decisions for a limited set of flexible workloads. This mode is opt-in, defaults to dry-run, requires a signed policy to go live, supports an immediate kill switch, and logs every action for audit. Latency-hard workloads are excluded.

Phases

The pilot is organized as a sequence of phases gated by evidence, not by a fixed calendar. Each phase is reversible and produces an artifact the customer can review independently before agreeing to proceed.

Phase	Activity	Gate to next phase
Onboarding	Ingest trace; confirm regions and SLA constraints	Trace validates; regions mapped
Offline replay	Produce historical-replay projection	Projection reviewed with customer
Shadow validation	Record live decisions; realize against settlement	Realized savings meet threshold
Controlled execution	Act on flexible workloads under policy	Operational sign-off

Success metrics

A pilot is evaluated on evidence meaningful to infrastructure and finance reviewers: realized cost reduction versus the current-price-only baseline, measured on the customer's own workload in shadow mode; agreement between predicted and realized savings, which indicates how dependable the projection is; zero SLA violations or deadline misses introduced by optimized placement; and bounded downside events, which the safety gate is designed to contain. The historical benchmark sets expectations; the shadow-mode result is the figure a pilot is judged on.

Security and data handling

The pilot's minimum footprint is a workload trace and read-only market-data access. Aurelius does not take custody of workloads, model weights, or data; in replay and shadow modes it does not connect to the customer's scheduler at all. Market-data credentials are read-only, no secrets are stored in the system, and controlled execution requires explicit, signed authorization. For controlled execution, the scheduler integration requires only the permission to submit and label jobs in the regions and queues explicitly in scope — the narrowest role that lets the agreed workloads be placed.

A formal compliance program (SOC 2) is on the roadmap and not yet in place. A pilot can be conducted entirely in read-only modes, which avoids granting any execution privilege while the compliance posture matures.

Known limitations entering a pilot

- Validated savings cover U.S. markets (CAISO, PJM, ERCOT); European markets require a connection that is implemented but not yet validated.
- Carbon-aware optimization currently has marginal-emissions coverage for CAISO only on the available data plan.
- Tier 2 and Tier 3 have been exercised against synthetic fixtures; live validation depends on customer-supplied queue and telemetry data.
- Controlled-execution adapters are unit-tested but not yet validated against live production infrastructure; first controlled execution should be scoped narrowly and reviewed jointly.

Appendix A — Workload trace format

The workload trace is a simple tabular export. Four fields are sufficient to begin; additional fields refine the optimization and can be supplied incrementally.

Field	Required	Description
Job identifier	Yes	Unique reference for the job
Workload type	Yes	Training, inference, batch, maintenance, etc.
Submission time	Yes	When the job became eligible to run
Duration	Yes	Estimated runtime
GPU count	Optional	Per-class default applied if omitted
Deadline / max delay	Optional	Scheduling flexibility window
Allowed / forbidden regions	Optional	Placement constraints
SLA class	Optional	Best-effort, standard, or guaranteed

Engineering-level setup, command-line usage, and reproduction commands are maintained separately in the developer documentation and are omitted here by design.